Ram Spectral Certificates for the Two-Dimensional $O(3)$ Particle Pole

Conditional Subthreshold Extensions and Controlled Moment Tests

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M388

October 8, 2026

Abstract

We analyze the source-specific argument for an isolated particle pole in the two-dimensional $O(3)$ spin field in OpenAI's October 4, 2026 manuscript. The reconstruction separates its assumed continuum construction from its transfer, parity, normalized-cluster visibility, state-counting, and positive mass-support arguments. Actual Ram Calculus passages on physical Gram metrics, full residuals, and finite positive moment records support a controlled reformulation of its spectral tests. Conditionally on the same uniform inputs, the state count extends from the displayed energy $3m_*/2$ to every fixed energy strictly below $2m_*$, and the field-visible mass support is finite on every compact mass interval below that threshold. We prove quantitative noisy-moment window bounds, full-residual visibility certificates, and residue enclosures when a pole and a rest-spectrum gap have independently been established. None of these finite certificates constructs the continuum field, proves the upstream renormalization-group estimates, or resolves the threshold itself. Relevance is L4/5 and is distinct from realized contribution. Contribution assessment: C2/5, the results give limited conditional subthreshold extensions and explicit local error certificates using classical tools without closing the source's global dependencies.

Contents

1 Question, Scope, and Logical Responsibility **2**

2 Spaces, Normalizations, and Unresolved Quantifiers **3**
2.1 The fixed lattice prescription 3
2.2 Mass support and operator domains 3

3 From Square Defects to Transfer Control **4**
3.1 Why rectangular control follows 4
3.2 Bounded insertions and normalized fields 5

4 Parity and Visibility: What Moments Must Preserve **5**
4.1 The quadratic covariance mechanism 5
4.2 An even-sector gap and a bounded detector 6

5 Normalized Clusters and the Required Limit Order **7**

6 Positive Cutoffs and a Subthreshold Extension **7**
6.1 The heat-product identity and its error 8
6.2 A polynomial with a summable high-energy tail 8

7 Mass-Support Reformulation and Extension **9**
7.1 Positive measures at fixed momentum 9

8 Ram Moment Coordinates With Explicit Error Budgets **11**
8.1 What the corpus actually permits 11
8.2 Physical Gram coordinates and omitted residuals 12
8.3 Pole residue after, not before, isolation 12

9 Current Primary Literature and Source Status **13**

10 Comparison, Advantages, and Remaining Obstacles **13**

11 Verification, Reproducibility, and Limits **14**

12 Future Research With Explicit Targets **15**

13 Conclusion and Contribution Assessment **16**

1 Question, Scope, and Logical Responsibility

The assigned manuscript is *An Isolated Particle Pole for the Two-Dimensional $O(3)$ Spin Field*, hereafter the source [1]. The analysis fixes the public source snapshot

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

An applicability score concerns the usefulness of a method at a stated step. It is not a verdict on validity, a proof of a new physical theory, or evidence of historical priority. The same distinction applies to the contribution score. A coordinate change that preserves an equation is not credited as a solution of its unproved premises.

The source claims an isolated positive atom in a Källén–Lehmann measure for one particular continuum limit. It does not claim that every massive $O(3)$-invariant field has an atom. A positive measure with continuous density on $[1, 4]$ has a positive mass-squared threshold and no atoms. Thus exponential decay or a finite transfer eigenvalue cannot alone answer the question. The source's uniform finite-volume estimates and visibility mechanism do essential work.

All seven assigned mathematical sections, and the source preamble and bibliography, were read in the author work. The parent separately read all seven sections and every displayed proof therein. The completed independent review examined the target manuscript and selected companion passages, not the full companion construction. It found no established blocking local error within that bounded scope. The continuum construction, its exact renormalization-group estimates, and its uniform source-control premises remain imported, not independently certified. This report proves the local additions explicitly and states exactly where those imported premises are used.

The private corpus was searched by local inventories and targeted actual passages. This is not a claim to have checked every technical report in the corpus. In particular, the present argument uses the physical-metric and finite-moment statements actually read in Volume 21, and the finite Taylor-remainder theorem in Volume 1 [3, 4, 5]. The public literature search used public topics and titles only; confidential source contents were not uploaded.

2 Spaces, Normalizations, and Unresolved Quantifiers

2.1 The fixed lattice prescription

Let σ be probability area measure on S^2. The cutoff lattice has spacing $a_N = L^{-N}$, where the prescribed L is a fixed dyadic integer. For a periodic nearest-neighbor graph Λ, its spin law is

$$Z_{N,\Lambda}^{-1} \exp\left(\beta_N \sum_{\{x,y\}\in E(\Lambda)} q_x \cdot q_y\right) \prod_{x\in\Lambda} \mathrm{d}\sigma(q_x), \qquad q_x \in S^2. \tag{2.1}$$

Each unoriented bond occurs once and there is no external field. The prescribed trajectory has $\beta_N = H + (\log L)N/(2\pi) + O(\log(N+1))$, with sufficiently large fixed terminal parameter H, and a prescribed positive normalization B_N. The normalized smeared component is

$$X_f^i = B_N a_N^2 \sum_x f(x) q_x^i, \qquad f \in C_c^\infty(\mathbb{R}^2), \quad i = 1,2,3. \tag{2.2}$$

All lengths are physical, not counts of lattice sites. The plane state is the periodic-square limit at fixed cutoff. This excludes changing to an arbitrary boundary state while retaining the source estimates.

The source's Proposition 2.1 imports three distinct statements from the companion continuum construction [2]:

H1. Plane joint moments of (2.2) converge to tempered, Euclidean-invariant, $O(3)$-covariant Schwinger distributions. Their Osterwalder–Schrader reconstruction has a unique vacuum Ω, a positive Hamiltonian gap $H_{\mathrm{phys}}|_{\Omega^\perp} \geq m > 0$, and a nonzero one-field reflection norm for a test strictly in positive time.

H2. At dyadic widths $w_k = M_0 2^k$, uniformly in all $N \geq K_0$,

$$0 \leq 4\log Z_N(w_k, w_k) - \log Z_N(2w_k, 2w_k) \leq Ce^{-cw_k}. \tag{2.3}$$

H3. Component moment measures of the normalized field are nonnegative and obey $C^r r!$ bounds on products of arbitrarily translated unit boxes, uniformly in cutoff and the prescribed square tori or plane. The uniformity includes separate translations of the boxes and each fixed finite list of component indices.

These are hypotheses for the deductions below, not three equivalent formulations of a mass gap. Their regulator and volume uniformities cannot be inferred from one finite numerical experiment.

2.2 Mass support and operator domains

The two-point function has the representation

$$W_2^{ij}(x) = \delta_{ij} \int_{[0,\infty)} \Delta_+(x;s)\rho(\,\mathrm{d}s), \qquad \mathcal{M} = \{\sqrt{s} : s \in \operatorname{supp}\rho\}, \quad m_* = \inf \mathcal{M}. \tag{2.4}$$

The measure variable is mass squared; m_* is mass. Symmetry makes the one-field vectors perpendicular to the vacuum. If ρ had support below m^2, a compact portion of that support and sufficiently small spatial momenta would give one-field energy below the Hamiltonian gap. The nonzero reflection norm excludes $\rho = 0$. Hence $0 < m \leq m_* < \infty$.

At positive Euclidean time,

$$C(t,b) = \int \rho(\,\mathrm{d}\mu^2) \int_{\mathbb{R}} \frac{\mathrm{d}p}{2\pi} \frac{e^{ipb - t\sqrt{p^2+\mu^2}}}{2\sqrt{p^2+\mu^2}}. \tag{2.5}$$

Positivity and temperedness imply polynomial growth of ρ on compact mass-squared intervals growing to infinity: restrict the momentum integral to $|p| \le 1$ and bound the shell weight from below by a multiple of $(1+R)^{-1}$ for $\mu \le R$. Positive time supplies exponential damping. Euclidean rotation of a separation of length r to $(r,0)$ then gives $0 < C(t,b) \le C_0 e^{-m_* r}$ for $r \ge 1$, by factoring out $e^{-m_* r}$ and dominating the remaining integral by its $r = 1$ value. No atom is presumed in this argument.

At fixed N, w, the row Hilbert space is $\mathcal{H}_{N,w} = L^2((S^2)^{w/a_N}, \sigma^{\otimes w/a_N}; \mathbb{C})$. The normalized transfer operator $T_{N,w}$ is bounded, positive, self-adjoint, compact and injective. Its generator $A_{N,w} = -a_N^{-1} \log T_{N,w}$ has spectral domain

$$D(A_{N,w}) = \{u : \sum_j E_j^2 |\langle \psi_j, u\rangle|^2 < \infty\}, \qquad T_{N,w}\psi_j = e^{-a_N E_j}\psi_j.$$

It is unbounded even though the transfer operator is bounded. A finite trial function must belong to this domain before a generator residual is used; a residual for T itself needs no logarithmic domain. There is no gauge constraint beyond the spin symmetry and the choice of vacuum complement for a gap calculation.

3 From Square Defects to Transfer Control

The transfer kernel puts half the horizontal interaction on each end:

$$K(q,q') = \exp\left\{ \frac{\beta_N}{2} \sum_x q_x \cdot q_{x+a} + \beta_N \sum_x q_x \cdot q'_x + \frac{\beta_N}{2} \sum_x q'_x \cdot q'_{x+a} \right\}. \tag{3.1}$$

The expansion of each crossing-bond exponential into tensor powers has positive coefficients and positive semidefinite kernels. Its finite diagonal integral proves trace class. A vector in the null space would, after multiplication by a bounded positive half-row weight, be orthogonal to all row-coordinate polynomials. Their density on the compact row space excludes such a vector. The strictly positive continuous kernel has a simple positive top eigenfunction. Divide by the top eigenvalue λ_0 to get T, with vacuum energy $E_0 = 0$ and $e_N(w) = E_1 > 0$. Define

$$R_{N,w}(t) = \operatorname{Tr} T_{N,w}^{t/a_N} = 1 + s_{N,w}(t), \quad s_{N,w}(t) = \sum_{j \ge 1} e^{-tE_j}, \quad \mathcal{N}_{N,w}(u) = \#\{j : E_j \le u\}.$$

The count includes the vacuum and multiplicities. Allowed times are lattice multiples; integer physical times are allowed since a_N^{-1} is integral.

3.1 Why rectangular control follows

Write $P_N(t,w) = (tw)^{-1} \log Z_N(t,w)$. At fixed w, this is $(a_N w)^{-1}$ times the logarithm of the ℓ^{t/a_N} norm of the eigenvalue sequence of K. It decreases with t; rotation gives separate monotonicity in w. Thus both lengths tending to infinity give $P_{\infty,N} = \inf_{t,w} P_N(t,w)$, finite at each cutoff. Equation (2.3) gives the exact telescoping difference

$$P_N(w_k, w_k) - P_N(2w_k, 2w_k) = \Delta_N(w_k)/(4w_k^2).$$

Summation and comparison with the largest dyadic square below $\min(t,w)$ yield, uniformly in N,

$$\begin{aligned} 0 \le P_N(t,w) - P_{\infty,N} &\le Ce^{-c\min(t,w)}, \\ 0 \le g_N(w) := a_N^{-1}\log\lambda_0(w) - wP_{\infty,N} &\le Cwe^{-cw}, \\ 0 \le \log R_{N,w}(t) &\le Ctwe^{-c\min(t,w)}. \end{aligned} \tag{3.2}$$

For the second line take $t\to\infty$; for the third use $\log R = tw(P_N - P_{\infty,N}) - tg_N(w)$ and $g_N \ge 0$. At $t = w$, $e^{-we_N(w)} \le s_{N,w}(w) \le C'w^2e^{-cw}$ for large w. Therefore $e_N(w) \ge c/2 > 0$ at every sufficiently large allowed width, uniformly in cutoff. This conclusion holds beyond dyadic widths and is needed when time and space are exchanged later.

3.2 Bounded insertions and normalized fields

A bounded slab observable F of time thickness r defines a normalized insertion operator A_F with $|A_F|$ pointwise dominated by $\|F\|_\infty T^{r/a}$. Hence $\|A_F\| \le \|F\|_\infty$. Subtract the vacuum term from $\mathrm{Tr}(A_F T^{(t-r)/a})/(1+s_{N,w}(t))$ to obtain

$$|\mathbb{E}_{\mathrm{tor};t,w}F - \langle\Omega, A_F\Omega\rangle| \le \|F\|_\infty [s_{N,w}(t-r) + s_{N,w}(t)]. \tag{3.3}$$

If the support has a lift of extent at most $3w/4$ in each direction, compare (w,w) to $(2w,w)$ using the same infinite-time cylinder, then to $(2w,2w)$ using the spatial transfer. Iterate and sum the exponentially small errors. This gives $|\mathbb{E}_{\mathrm{tor};w,w}F - \mathbb{E}_{\mathrm{plane}}F| \le C\|F\|_\infty e^{-cw}$, uniformly in cutoff. The bound for every bounded measurable local F also supplies total-variation convergence of the corresponding local marginals; weak convergence alone would not justify this use for sign events.

The factorial moment input gives, for fixed test shapes and arbitrary translations, $\mathbb{E}|X_f|^r \le A^r r!$ and $\mathbb{E}e^{\kappa|X_f|} \le C$ for small $\kappa > 0$. Even moments follow by replacing test factors with their absolute values against the nonnegative component moment measures; Cauchy–Schwarz gives odd absolute moments. The same argument yields $\|X_f - X_g\|_{L^r} \le C_r\|f-g\|_\infty$ on a common bounded support. To compare unbounded field products, truncate each X at $\pm w$. Their product is bounded by a fixed power of w, absorbed by the exponential comparison. Hölder and the exponential tails control the removed parts. Normalized field correlations thereby have uniform exponential torus-to-plane errors. Ignoring B_N in this step would be an unjustified change of observable.

4 Parity and Visibility: What Moments Must Preserve

4.1 The quadratic covariance mechanism

On any fixed finite ferromagnetic S^2 spin graph, the source proves for even-degree component monomials P, Q with site lists $\mathcal{A}, \mathcal{B}$ that

$$|\operatorname{Cov}(P,Q)| \le C_{r,s}\left(\max_{x\in\mathcal{A},y\in\mathcal{B}} \mathbb{E}q_x^3 q_y^3\right)^2. \tag{4.1}$$

Repeated sites remain distinct factors. The constant depends on the degrees, not the graph size. The reasoning is important because a formal covariance matrix alone does not establish (4.1). Write $q = (\cos\alpha\cos\vartheta, \cos\alpha\sin\vartheta, \sin\alpha\, u)$ with $0 \le \alpha \le \pi/2$ and $u \in \{-1,1\}$. Conditionally on α, the signs and rotators are independent, with couplings $J_{xy}\sin\alpha_x\sin\alpha_y$ and $J_{xy}\cos\alpha_x\cos\alpha_y$. The former increase and the latter decrease with the angles. For either model,

$$\partial_x\partial_y \log Z = \sum_e \mathbb{E}O_e\,\partial_x\partial_y J_e + \sum_{e,f}\operatorname{Cov}(O_e,O_f)\partial_x J_e\partial_y J_f \ge 0.$$

GKS and Ginibre positivity apply to the bond observables, and products of two same-sign derivatives are nonnegative. Thus the marginal angle law is associated. Like-component even-monomial covariances are nonnegative; unlike-component ones are nonpositive, by conditional independence and opposite monotonicities.

For clarity, cosine covariance positivity in the rotator model has its own replica proof. Set $\vartheta = U + V$, $\vartheta' = U - V$ in two independent copies; this surjective torus homomorphism preserves Haar integration. An observable difference is $-2\sin(k \cdot U)\sin(k \cdot V)$. Expansion of the replicated bond exponential gives nonnegative coefficients times squares of real integrals. This proves the required covariance sign, without assuming independence of neighboring spins.

The uniform sphere moment tensor is the sum over pairings divided by $3 \cdot 5 \cdots (2h+1)$ at degree $2h$. Expanding the interactions and contracting internal lines gives the external moment as $\sum_\pi d_\pi \prod_{\{b,c\}\in\pi} \mathbf{1}_{i_b = i_c}$ with $d_\pi \geq 0$. A pairing crossing two even lists has at least two crossings. Color these two pairs differently and all other factors with the third color. The component sign comparison bounds its coefficient by the product of two across-list correlations, hence by G^2. Comparing same-color and different-color assignments then controls single-component covariances. If an individual monomial is odd in some color, its mean is zero and each compatible joint pairing crosses, giving the same bound.

The remaining mixed-even case needs an additional argument rather than merely subtracting pairings. Write the conditional mean as $a_P b_P$, where a_P is the color-3 sign mean including amplitudes; replace the other colors by color 1 to get a decreasing positive comparison B_P. The mixed Fourier coefficients have absolute values bounded by the all-cosine coefficients. Consequently $B_P \pm b_P$ are decreasing, a_P is increasing, and $D_P \pm a_P b_P$ are increasing for $D_P = a_P - B_P$. Association, applied to the four choices of signs, yields

$$|\operatorname{Cov}(a_P b_P, a_Q b_Q)| \leq \operatorname{Cov}(D_P, D_Q).$$

Expanding the right side reduces it to the already bounded like- and unlike-color covariances. Conditional covariances are controlled by the same Fourier domination. The total covariance identity proves (4.1). This is the source's argument; it is not claimed as a new Ram theorem.

4.2 An even-sector gap and a bounded detector

Set $\tau = e^{-a e_N(w)} \in (0,1)$. Across j rows, a component two-point correlation is at most τ^j. For a centered even row polynomial vector v_P, (4.1) gives $0 \leq \langle v_P, T^j v_P \rangle \leq C_P \tau^{2j}$. The spectral measure is positive, so it has no support above τ^2: positive mass above that value would violate the inequality as $j \to \infty$. Even row polynomials are dense, and the positive continuous vacuum function is bounded away from zero at fixed N, w. Thus the even vacuum-complement norm of T is at most τ^2. Compactness gives a top nonvacuum eigenfunction ψ; since $\tau > \tau^2$, it is odd. Commuting component reflections allow a real unit eigenfunction odd in a selected component.

Let $b = \langle \Omega, |\psi| \rangle^2$. Kernel positivity and the even-sector bound imply $\tau \leq \langle |\psi|, T|\psi| \rangle \leq b + (1-b)\tau^2$, hence

$$b \geq \frac{\tau}{1+\tau} \geq \frac{\tau}{2}. \tag{4.2}$$

The detector $F = \operatorname{sgn}\psi$, with $\operatorname{sgn} 0 = 0$, is bounded and component-odd. Its existence at fixed cutoff is not yet survival of a particle after the limits.

5 Normalized Clusters and the Required Limit Order

Conditionally on the angles, Edwards–Sokal bonds couple the selected Ising signs. Their law is associated and increases with the angle array; a second use of association gives joint association of angles and bonds. Let $d = \lfloor \sqrt{w} \rfloor$ in physical units. The positive spectral expansion of the torus correlation of F contains the ψ, Ω term, so (4.2) gives

$$\mathbb{P}_{\mathrm{tor}}(\text{a sign cluster meets rows } 0, d) \geq c e^{-(d+a)e_N(w)}.$$

Off the connection event, flipping every cluster meeting the first row preserves the conditional law and reverses only the first detector. This justifies the probability bound.

For nonnegative tests, the correct cluster weight is

$$x_{\mathcal{C}}(h) = B_N a^2 \sum_{y \in \mathcal{C}} h(y) \sin \alpha_y, \qquad S(h, h') = \sum_{\mathcal{C}} x_{\mathcal{C}}(h) x_{\mathcal{C}}(h'). \tag{5.1}$$

Conditional spin signs give $\mathbb{E}(X_h X_{h'} \mid \alpha, \eta) = S(h, h')$. Both increasing angles and merging clusters increase S. For two fixed separated test shapes the positive continuum kernel bounds $\mathbb{E}S$ below, and the uniform fourth moments bound $\mathbb{E}S^2$ above. The second-moment inequality gives $\mathbb{P}\{S \geq \epsilon\} \geq c_0 > 0$, uniformly at large cutoff and large physical width. This is a statement about normalized spin weight, not merely a geometric connection probability.

Such a connection supplies a short-direction crossing of an annular strip. Transfer to the cutoff plane, use the square-symmetric positive-association crossing theorem [6], and transfer back. Four long-direction strip crossings form a connected separating barrier in a local torus chart. The theorem is applied on the plane, not directly to a torus with unverified hypotheses. It needs neither independence nor an assumed spatial Markov property. Association on cylinder events passes to the plane; compact-set separation by finite increasing cylinder events extends it to increasing Borel events.

On the barrier event all clusters linking the two local tests merge into its cluster. For row centers s, t, a path between the row intervals plus the two increasing weighted barrier events therefore gives $\mathbb{E}X_{H_s}X_{H_t} \geq c\mathbb{P}(D_{st})$ for the fixed enlarged test shapes H_s, H_t. Covering each row by $O(w)$ intervals, and comparing each pair with the plane, yields

$$\limsup_{N \to \infty} e^{-(d+a)e_N(w)} \leq C w^2 (e^{-m_* d} + e^{-cw}). \tag{5.2}$$

For each fixed width there are finitely many translates, so plane moment convergence may be applied first. If $L_w = \liminf_N e_N(w) < \infty$, take a subsequence attaining it; then $a e_N(w) \to 0$ on that subsequence. Taking logarithms in (5.2) and then letting w grow proves

$$\liminf_{k \to \infty} \liminf_{N \to \infty} e_N(w_k) \geq m_*. \tag{5.3}$$

If $L_w = \infty$, the lower bound is automatic. For every $v < m_*$, all sufficiently large fixed w_k thus have $e_N(w_k) > v$ for all sufficiently large N depending on that width. The proof does not give a single cutoff threshold valid at every width.

6 Positive Cutoffs and a Subthreshold Extension

The source proves a linear count at $U_0 = 3m_*/2$. Its proof actually allows any fixed threshold strictly below $2m_*$, provided all constants and the base width are allowed to depend on its distance from that endpoint. This is a parameter extension of the same mechanism, not a replacement for visibility or for the input construction.

6.1 The heat-product identity and its error

Rotation of the same torus partition function gives

$$e^{tg_N(w)}R_{N,w}(t) = e^{wg_N(t)}R_{N,t}(w).$$

Choose $D = \lfloor\sqrt{w}\rfloor$ and $t_0 = \lfloor\sqrt{D}\rfloor$. For integer $t_0 \le t \le D$,

$$\begin{aligned}\log R_{N,2w}(t) - 2\log R_{N,w}(t) = t(2g_N(w) - g_N(2w)) \\ + \log R_{N,t}(2w) - 2\log R_{N,t}(w).\end{aligned}$$

The first term is exponentially small by (3.2). The uniform gap at every sufficiently large circumference gives $s_{N,t}(w) \le e^{-c_*(w-t)}s_{N,t}(t)$, and the last factor is uniformly bounded. Also (3.2) bounds the logarithms of the two original traces by a polynomial in w times e^{-ct_0}. Exponentiation therefore gives the absolute, not just relative, bound

$$|R_{N,2w}(t) - R_{N,w}(t)^2| \le Ce^{-cw}. \tag{6.1}$$

Uniformity in N is inherited from (2.3) and cannot be replaced by an estimate at one cutoff.

6.2 A polynomial with a summable high-energy tail

For y in a fixed compact interval of positive energies, set

$$b_w = D^{-1/8}, \qquad j_0 = \lceil De^{-(y+b_w/2)}\rceil, \qquad Q_{w,y}(x) = \sum_{j=j_0}^{D}\binom{D}{j}x^j(1-x)^{D-j}.$$

It is a binomial tail, so $0 \le Q \le 1$ on $[0,1]$. For large w, uniformly over the specified y interval,

$$\begin{aligned}&Q(e^{-E}) \ge 1 - Ce^{-cD^{3/4}} && (0 \le E \le y),\\ &Q(e^{-E}) \le Ce^{-cD^{3/4}}e^{-t_0E} && (E \ge y + b_w).\end{aligned} \tag{6.2}$$

Indeed the threshold fraction differs from the two endpoint probabilities by at least cb_w after allowing for rounding $1/D$. Hoeffding's inequality gives $e^{-cDb_w^2}$ [9]. To prove the extra tail factor, choose $j_0 \ge c_1D$. On a fixed bounded energy interval the loss e^{t_0E} is absorbed by $e^{-cD^{3/4}}$ since $t_0 = O(D^{1/2})$. Beyond a sufficiently large fixed E_*, use $Q(e^{-E}) \le 2^De^{-j_0E}$ and $j_0 - t_0 \ge c_1D/2$. The resulting exponential in DE proves the second inequality for the entire unbounded energy tail.

Every nonzero monomial degree of Q lies in $[j_0, D] \subset [t_0, D]$. The sum of absolute monomial coefficients is at most $\sum_{j=j_0}^{D}\binom{D}{j}2^{D-j} \le 3^D$. Applying (6.1) at each monomial gives

$$\left|\sum_{E \text{ at } 2w} Q(e^{-E}) - \sum_{E,E' \text{ at } w} Q(e^{-(E+E')})\right| \le Ce^{-c'w}. \tag{6.3}$$

Here 3^D is absorbed because $D = o(w)$. The positive minimum degree ensures absolute convergence of the infinite spectral sums. A cutoff polynomial uniformly small at high energy but lacking the factor e^{-t_0E} would not suffice for an infinite state count.

Theorem 6.1 (Every fixed energy below two excitations)**.** *Assume H1–H3, the transfer consequences in Section 3, and visibility* (5.3)*. For every fixed U with $0 < U < 2m_*$ there exist $C_U < \infty$ and a dyadic base index k_U such that*

$$\limsup_{N\to\infty} \mathcal{N}_{N,w_k}(U) \le C_U w_k \quad (k \ge k_U). \tag{6.4}$$

There is no asserted uniformity of C_U, k_U as $U \uparrow 2m_*$.

Proof. Fix numbers $U < U_1 < V < 2m_*$. By visibility, for every sufficiently large fixed dyadic w, all sufficiently large cutoffs have $2e_N(w) > V$. Increase the base width so that $U_1 + b_w < V$. For $y \in [U, U_1]$, every pair with $E + E' \le y + b_w$ has a vacuum component, since two nonvacuum energies are greater than V. There are exactly $2\mathcal{N}_{N,w}(y + b_w) - 1$ such pairs; replacing this by twice the count is an upper bound. The high-energy pairs in (6.3) contribute at most $Ce^{-cD^{3/4}} R_{N,w}(t_0)^2 \le Ce^{-cD^{3/4}}$ by (6.2). The low-energy sum at $2w$ is at least $(1 - Ce^{-cD^{3/4}})\mathcal{N}_{N,2w}(y)$. Absorb the additive errors using $\mathcal{N} \ge 1$, to get

$$\mathcal{N}_{N,2w}(y) \le 2(1 + \epsilon_w)\mathcal{N}_{N,w}(y + b_w), \qquad \epsilon_w \le Ce^{-c\lfloor\sqrt{w}\rfloor^{3/4}}. \tag{6.5}$$

The constants may depend on U, U_1, V, but not on N, w, y. At fixed width the cutoff threshold may depend on that width.

Choose a dyadic base w_{k_U} large enough that $\sum_{k \ge k_U} b_{w_k} < U_1 - U$. Also $\prod_{k \ge k_U}(1 + \epsilon_{w_k}) < \infty$. At the base width, $\mathcal{N}_{N,w_{k_U}}(U_1) \le e^{w_{k_U} U_1} R_{N,w_{k_U}}(w_{k_U}) \le C_0$ uniformly in N. For a fixed target index k, iterate (6.5) backwards along its finite chain, starting at threshold U. Every shifted threshold stays below U_1. There is a common sufficiently large cutoff for this finite chain, and

$$\mathcal{N}_{N,w_k}(U) \le C_0 2^{k - k_U} \prod_{h=k_U}^{k-1} (1 + \epsilon_{w_h}) \le C_U w_k.$$

Take the cutoff limsup for that fixed k. No uniform cutoff threshold for infinitely many widths and no diagonal limit have been used. □

This theorem weakens the numerical restriction on a fixed energy window, not the hypotheses on the spin-field construction. At the endpoint $2m_*$ the exclusion of pairs of excited states disappears. Their count can have a different growth order. Neither this proof nor a change of moment coordinates establishes the endpoint.

7 Mass-Support Reformulation and Extension

7.1 Positive measures at fixed momentum

Choose nonzero nonnegative $g, h \in C_c^\infty$, with $\operatorname{supp} g \subset (0, l)$ strictly away from zero. For $f(t, b) = g(t)h(b)$, truncate X_f at $\pm w$ and form $v = A_F\Omega$ with the left slab variable at time zero. The first insertion is then reflected into $[-l, 0]$, the other lies in $[n, n + l]$. For the untruncated linear insertion, $A_{X_f}\Omega = a\sum_t g(t)T^{t/a}Q_{h,N}\Omega$. This fixes the sign of the time damping in the limiting measure.

Let U_b be spatial row translation and use the momentum projection $\Pi_p = (a/w)\sum_b e^{-ipb}U_b$, where $p \in (2\pi/w)\mathbb{Z} \cap [-\pi/a, \pi/a)$. Define

$$\nu_{N,w,p} = w \sum_E \|P_E \Pi_p v\|^2 \, \delta_{e^{-E}} \quad \text{on } [0, 1]. \tag{7.1}$$

This is a finite positive measure. The factor w is essential: its nth moment is the spatial Riemann sum $a\sum_b e^{-ipb}\langle v, T^{n/a}U_b v\rangle$. Replacing w by one would change the state-count comparison.

Let $G(E) = \int_0^l g(t)e^{-tE}\,\mathrm{d}t$ and $\widehat{h}(p) = \int e^{-ipb}h(b)\,\mathrm{d}b$. The limiting measure ν_p is the pushforward under $\mu \mapsto e^{-\sqrt{p^2+\mu^2}}$ of

$$W_p(\mu)\rho(\,\mathrm{d}\mu^2), \qquad W_p(\mu) = \frac{|\widehat{h}(p)|^2 |G(\sqrt{p^2 + \mu^2})|^2}{2\sqrt{p^2 + \mu^2}}. \tag{7.2}$$

For each fixed integer $n \geq 0$ and compact momentum interval I, the source proves

$$\lim_{k\to\infty} \limsup_{N\to\infty} \max_{p\in I} \left| \int x^n \, \mathrm{d}\nu_{N,w_k,p} - \int x^n \, \mathrm{d}\nu_p \right| = 0. \tag{7.3}$$

The maximum is over allowed lattice momenta. To track its uniformity, the support extents of the two slab observables are less than $3w/4$ for large fixed w; their supremum product is $O(w^2)$. Torus/cylinder/plane comparison and tails leave an exponentially small error even after summing $O(w/a)$ translations with weight a. At fixed w, the difference-test bound gives equicontinuity of translated cutoff correlations, so pointwise moment convergence upgrades to uniform convergence on that compact translation interval. The compact momentum interval gives uniform Riemann-sum convergence. The remaining spatial tail is exponentially small by (2.5). The zeroth moment is valid: reflected slab supports are still separated because g is supported strictly in positive time.

The zeroth moment supplies a uniform mass bound in this iterated limit. Approximation by polynomials in x with coefficients continuous in p then upgrades (7.3) to every $\Psi \in C(I \times [0,1])$. In particular, no mass at the compactified high-energy point $x = 0$ is silently discarded. This is a use of positive moment determination on a specified compact support, not a finite-atom reconstruction from finitely many exact moments.

Theorem 7.1 (Finite support throughout compact subthreshold windows)**.** *Under the hypotheses of Theorem 6.1 and* (7.3)*, for every fixed M with $m_* \leq M < 2m_*$,*

$$\mathcal{M} \cap [m_*, M] \quad \textit{is finite.}$$

Thus each mass in $[m_*, 2m_*)$ is isolated in $\mathcal{M}$ and carries a positive finite atom. Infinite accumulation at $2m_*$, or above it, is not excluded.

Proof. Choose $M < U < 2m_*$. Take $p_0 > 0$ so small that $\sqrt{M^2 + p_0^2} < U$ and $\widehat{h}(p) \neq 0$ on $I = [-p_0, p_0]$; this is possible since $\widehat{h}(0) > 0$. This interval is fixed independently of the number of selected masses. For any finite collection of J distinct masses μ_j in the indicated support, their curves $x_j(p) = e^{-\sqrt{p^2+\mu_j^2}}$ are disjoint compact graphs in $I \times (e^{-U}, 1)$. Choose nonnegative continuous windows Ψ_j equal to one on these graphs, with mutually disjoint neighborhoods inside that strip. The support condition on ρ, positivity of $G(E)$, and nonvanishing of $\widehat{h}$ imply $L_j(p) = \int \Psi_j(p,x) \, \mathrm{d}\nu_p(x) > 0$ for every p. Continuity by exponentially dominated integration and compactness give $\eta = \min_{j,p} L_j(p) > 0$. Both η and required approximation scales may depend on the finite collection.

Continuous-test convergence gives positive window integrals for all sufficiently large fixed dyadic w and then all sufficiently large N, at every allowed $p \in I$. Each window requires an energy below U in that momentum sector. Distinct windows at the same momentum require distinct energies, and distinct momentum sectors are orthogonal. For large w there are at least $c_0 w$ allowed momenta in I, with $c_0 > 0$ independent of J. Hence $J c_0 w \leq \mathcal{N}_{N,w}(U)$. Take the cutoff limsup at fixed w and use Theorem 6.1, obtaining $J \leq C_U/c_0$. Arbitrarily large finite subsets are impossible, so the support is finite. The support minimum is $m_* > 0$ and the measure is locally finite. An isolated support point must have positive mass, and that mass is finite. For any point strictly below $2m_*$ choose a larger M still below it to obtain isolation on both sides. □

Taking just $M > m_*$ recovers the source conclusion $\rho = Z\delta_{m_*^2} + \rho_{\mathrm{rest}}$ with $0 < Z < \infty$ and $\operatorname{supp} \rho_{\mathrm{rest}} \subset [(m_* + \delta)^2, \infty)$ for some $\delta > 0$. The source used $M = 5m_*/4$ and $U = 3m_*/2$; the theorem removes those convenient fixed fractions. It does not compute m_*, Z, δ, the number of subthreshold masses, or a scattering matrix. It concerns the spectrum visible to this spin field, not all sectors of the continuum Hilbert space.

8 Ram Moment Coordinates With Explicit Error Budgets

8.1 What the corpus actually permits

Volume 21's finite-moment chapter [4] distinguishes feasibility, determinacy, identified observables and stability. A Hankel matrix $H_d = (m_{i+j})$ is positive semidefinite for a positive measure, but positivity alone does not encode its support. On $[0,1]$ the localizers for x, $1-x$, and $x(1-x)$ supply additional necessary tests at the available degrees. A finite annihilator certificate requires an exactly zero quadratic form, real roots on the declared support, positive weights, and reproduction of every supplied moment. A numerically small eigenvalue is not an exact support equation.

These statements apply directly to (7.1) and (7.2). The variables are $x = e^{-E} \in [0,1]$ and positive field-dependent weights, not the unweighted energy count. Multiplicities enter the trace count separately. The moment coordinate map preserves polynomial integrals exactly, but it need not identify the number of mass poles from a finite record. Two positive measures can have the same low moments and different supports; the finite nullspace examples in [4] make this limitation explicit.

Volume 1's actual local Taylor theorem [5] is a finite identity plus an integral remainder, not an unconditional infinite differential expansion. Its one-dimensional measure specialization is

$$\int \psi(x)\,\mathrm{d}\nu(x) = \sum_{j=0}^{d} \frac{\psi^{(j)}(c)}{j!} \int (x-c)^j\,\mathrm{d}\nu(x) + \mathcal{R}_d, \quad |\mathcal{R}_d| \le \frac{\left\|\psi^{(d+1)}\right\|_\infty}{(d+1)!} \int |x-c|^{d+1}\,\mathrm{d}\nu(x). \tag{8.1}$$

For $\psi \in C^{d+1}([0,1])$ this follows by ordinary Taylor's theorem along the interval from $c \in [0,1]$ to x and integration. Its remainder is zero for polynomials of degree at most d. For merely continuous spectral windows Bernstein approximation is the appropriate alternative; differentiability should not be invented to apply (8.1). Neither representation asserts that the finite trial space is invariant under the transfer generator.

Proposition 8.1 (Noisy window transport)**.** *Let $\nu, \widehat{\nu}$ be positive measures on $[0,1]$, each of mass at most M_0. Suppose their moments through d differ by at most η_j. If $P(x) = \sum_{j=0}^{d} c_j x^j$ obeys $\|\Psi - P\|_\infty \le \epsilon$, then*

$$\left| \int \Psi\,\mathrm{d}\nu - \int \Psi\,\mathrm{d}\widehat{\nu} \right| \le 2M_0\epsilon + \sum_{j=0}^{d} |c_j|\eta_j. \tag{8.2}$$

The same assertion holds uniformly in a compact momentum parameter when every bound on the right is uniform in that parameter.

Proof. Subtract the two polynomial integrals and bound each coefficient error. Each replacement of Ψ by P costs at most $M_0\epsilon$. Add the three bounds. For a parameter-dependent polynomial take suprema of its coefficients; all sums are finite. □

In the source's count proof, monomial heat-trace errors δ_j therefore cost at most $\sum |c_j|\delta_j$, bounded by $3^D \max_j \delta_j$ for its binomial cutoff. This explicitly explains why exponentially small heat errors tolerate growing degree, whereas fixed-precision errors generally do not. In the mass-support proof, (8.2) smaller than $\eta/2$ is sufficient to keep a selected window positive. It supplies a stopping criterion only if the approximation and moment errors have certified upper bounds; an observed numerical residual is not automatically such a bound. The proposition does not improve the source's convergence rate unless rates for the moments are separately supplied.

For reference, Bernstein approximation also gives a transparent continuous-test bound. Let $B_d\Psi(x) = \mathbb{E}\Psi(Y/d)$ with $Y \sim \mathrm{Bin}(d, x)$ and modulus of continuity ω_Ψ. For any $\delta > 0$, Chebyshev and $\mathrm{Var}(Y/d) \le 1/(4d)$ give

$$\|B_d\Psi - \Psi\|_\infty \le \omega_\Psi(\delta) + \frac{\|\Psi\|_\infty}{2d\delta^2}. \tag{8.3}$$

Indeed split into $|Y/d - x| \le \delta$ and its complement, bounding the latter difference by $2\|\Psi\|_\infty$. Choose, for example, $\delta = d^{-1/4}$ to obtain convergence. For momentum-dependent windows the common modulus in x on the compact product is used. This classical argument is more than a formal statement that moments "determine" the limit: it exhibits the mass and approximation terms.

8.2 Physical Gram coordinates and omitted residuals

For injective synthesis $J : \mathbb{C}^d \to \mathcal{H}$, set $G = J^*J > 0$, $P = JG^{-1}J^*$, $K = J^*AJ$ and $B = G^{-1}K$, with columns of J in $D(A)$ if A is unbounded. The corpus result [3] gives

$$\|(A-\lambda)Jc\|^2 = \|(B-\lambda)c\|_G^2 + \|(I-P)AJc\|^2, \qquad \|c\|_G^2 = c^*Gc. \tag{8.4}$$

The proof is the orthogonal decomposition into P and $I - P$. For self-adjoint A, $GB = B^*G$ and $G^{1/2}BG^{-1/2}$ is Hermitian. A small coefficient residual in an untransported Euclidean metric does not give (8.4); even a zero compressed residual can omit all the physically relevant error.

Proposition 8.2 (Residual plus overlap gives a visible window)**.** *Let A be self-adjoint, $u \in D(A)$ unit, $\lambda \in \mathbb{R}$, $\|(A-\lambda)u\| \le \epsilon$, and $d > 0$. Write $P_I = \mathbf{1}_{[\lambda-d,\lambda+d]}(A)$. For every $v \in \mathcal{H}$,*

$$\|P_I v\| \ge \left(|\langle v, u\rangle| - \|v\|\ \epsilon/d\right)_+. \tag{8.5}$$

Proof. The spectral theorem gives $\|(I-P_I)u\| \le \epsilon/d$. Split the overlap into its orthogonal window and complement parts. The first is at most $\|P_I v\|\,\|P_I u\| \le \|P_I v\|$; the second is at most $\|v\|\,\epsilon/d$. Rearrange and use nonnegativity. □

This is a finite or infinite Hilbert-space statement with an actual full residual, not a compressed eigenvalue test. With $v = \Pi_p A_F \Omega$, it certifies positive field weight in a finite energy window if the overlap lower bound exceeds the remainder. For (7.1), the window weight is $w\,\|P_I v\|^2$. Such a certificate at one finite N, w does not survive the double limit without uniformity. Nor does it identify a single atom within the window. Those conclusions require the source's separate count and positive mass-support argument.

8.3 Pole residue after, not before, isolation

Proposition 8.3 (Moment enclosure of an already isolated weight)**.** *Suppose a positive measure on $[0,1]$ is known to have the form $\nu = z\delta_{x_*} + \nu_{\mathrm{rest}}$, where $0 \le r < x_* \le 1$, $\operatorname{supp}\nu_{\mathrm{rest}} \subset [0,r]$, and $\nu([0,1]) \le M_0$. For integer $k \ge 1$, let $|\widehat{m}_k - \int x^k\,\mathrm{d}\nu| \le \eta_k$. Then*

$$\max\left(0, \frac{\widehat{m}_k - \eta_k - M_0 r^k}{x_*^k - r^k}\right) \le z \le \min\left(M_0, \frac{\widehat{m}_k + \eta_k}{x_*^k}\right). \tag{8.6}$$

Intersect these intervals over any finite set of k.

Proof. Positivity gives $zx_*^k \le m_k$. The rest mass is at most $M_0 - z$, so $m_k \le zx_*^k + (M_0 - z)r^k$. Combine these two inequalities with $\widehat{m}_k \pm \eta_k$ and divide only by the strictly positive denominators. Intersection preserves every justified bound. If computed lower bounds exceed upper bounds, the proposed model or error budgets are inconsistent; one must not return a negative residue. □

In (7.2), if m_* and a rest gap $\delta > 0$ are independently certified, take $x_*(p) = e^{-\sqrt{p^2+m_*^2}}$ and $r(p) = e^{-\sqrt{p^2+(m_*+\delta)^2}}$. The enclosed weight is $z(p) = W_p(m_*)Z$, not Z itself. Where $W_p(m_*) > 0$, division transports the enclosure to Z. Uncertainty in m_*, δ, W_p requires interval bounds on those quantities as well; point estimates cannot simply be substituted. Increasing k reduces rest-spectrum leakage by $(r/x_*)^k$ but amplifies absolute moment errors by roughly x_*^{-k}. As the gap closes the lower-bound denominator deteriorates. These are useful controls on a proposed computation, not measurements of the actual $O(3)$ residue supplied by this report.

9 Current Primary Literature and Source Status

The public repository history checked on October 8, 2026 lists three withdrawals and revisions elsewhere in the collection [10]. A complete public preprint-subtree comparison between the pinned snapshot and commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb` found no changed blobs or current-edition links for the selected 100 manuscripts or the 59 direct companions explicitly mapped from their source links. The present manuscript and its massive-continuum companion remain identical in that comparison. This is an edition/provenance check, not a proof verification; unlisted and transitive mathematical dependencies are not certified by it.

Köhler-Schindler and Tassion's crossing theorem [6] supplies exactly the kind of scale-uniform comparison used by the source: square symmetries and positive association suffice for its planar crossing bound. The theorem statement and relevant definitions were checked in the independent review, not its entire proof. Its use is not replaced by a finite Ram moment calculation. Tokushige's 2025 primary article [7] gives a graphical proof of Ginibre's inequality for finite-graph XY models. Its stated theorem uses neutral integer charges; it supports the classical conditional-rotator ingredient, not the full mixed $O(3)$ covariance theorem. The source's replica and mixed-component arguments are therefore retained rather than attributed wholesale to that newer paper.

Balog and Niedermaier [8] develop form-factor spectral information within a massive-particle description. Their selected Section 2.1 uses an asymptotic multiparticle resolution, whereas the present source seeks an isolated pole for a specified lattice-limit field. Those are different starting assumptions. The form-factor framework can motivate subsequent residue or scattering studies, but it cannot discharge H1–H3 merely by physical agreement. Classical Osterwalder–Schrader reconstruction, GKS/Ginibre/FKG inequalities, Edwards–Sokal coupling, positive spectral calculus, and polynomial approximation retain their original attributions. No exhaustive literature search or priority claim is made here.

10 Comparison, Advantages, and Remaining Obstacles

Step	Source mechanism	Ram reformulation and actual boundary
Continuum field	H1–H3 from a specified massive construction.	Domains, normalization, and error prerequisites are made explicit; no new construction or RG estimate is proved.

Step	Source mechanism	Ram reformulation and actual boundary
Parity	Positive tensor pairings, conditional monotonicity, and dense even polynomial tests.	Moment positivity preserves the algebraic tests; mixed covariances and uniform physical limits still require the source arguments.
Visibility	Normalized cluster weights, circuits, bounded detector, and cutoff-first limits.	Full-residual and overlap certificates control a trial spectral window. They do not replace circuits or prove limiting visibility.
State count	Heat-product estimate and positive binomial cutoff at $3m_*/2$.	Theorem 6.1 allows each fixed $U < 2m_*$; coefficient-weighted moment errors are explicit. The endpoint remains outside the argument.
Mass support	Fixed momentum interval and positive windows give finitely many masses near the bottom.	Theorem 7.1 extends to every compact subthreshold mass window. This is not finiteness on the entire half-open interval.
Pole weight	Existence of a positive residue, no numerical residue computed.	Proposition 8.3 gives conditional enclosures from certified moments, location and rest gap; none of those numerical inputs is produced for the actual continuum model here.

The useful advantages are local and explicit. Moment coordinates separate observations from a proposed support model. Physical Gram metrics prevent coordinate-dependent residual claims. The noisy-window bound states how much precision a polynomial actually costs. The subthreshold extension shows that the source's fractions are convenient choices rather than intrinsic restrictions. Residue enclosures reveal a tradeoff between suppressing high-energy leakage and amplifying noise. These are transparent control mechanisms for future rigorous computations.

There are corresponding disadvantages. Moment inversion is ill conditioned as spectral points approach one another or the rest gap closes. A high-degree monomial expansion can have exponentially large coefficient norm. A nonorthogonal Ram representation can have an ill-conditioned Gram matrix, even when each entry is accurately computed. Finite positivity tests cannot replace a continuum operator domain or a uniform volume estimate. The local certificates may therefore be more expensive than a source-specific analytic argument and do not presently demonstrate a runtime advantage.

Strengthening hypotheses can make useful subproblems tractable: a known compact support, independently certified gap, uniformly bounded zeroth moment, quantitative moment errors, or an invariant finite trial space make the displayed enclosures implementable. Such assumptions must be declared, not counted as new solutions of the unconditional field problem. Weakening the numerical window restrictions was justified above; weakening reflection positivity, field normalization, association, or the regulator-uniform torus estimates has not been justified.

11 Verification, Reproducibility, and Limits

The separate source review records exact finite diagnostics using integers and rational arithmetic: 2,080 binomial cutoff polynomials of degrees at most 64, 18,720 rational evaluations, 84 even sphere moments through total degree 12, 4,950 rational overlap parameters, and nine finite vacuum-pair counting examples. They agree with the source algebra in their specified finite ranges. They do not prove the asymptotic estimates, RSW, the full finite-graph covariance inequality, the continuum

construction, or an isolated mass. All general claims in this report rely on their displayed proofs and declared analytic premises, not extrapolation from these tests.

A separate exact-rational extension check passed 6,348 noisy-window bounds over 529 measure pairs, 5,120 full-residual bounds, 76,800 field-overlap bounds, 21,600 exact pole-weight enclosures, and 129,600 noisy enclosures. These include 1,440 sharp-lower-bound equality cases, zero weights, saturated mass caps, and both noise signs. The checks also include 12 threshold/shift examples and 12 compact-band examples; these last arithmetic checks do not simulate the source's continuum state count. Two negative controls demonstrate why omitting the zeroth moment or replacing a full residual by a compressed residual is invalid. The independent reviewer read the actual proofs of Theorems 6.1–7.1 and Propositions 8.1–8.3; its bounded conclusion agrees with the displayed assumptions. Neither enumeration nor that local review supplies the missing upstream global verification.

The local audit files distinguish assigned-source reads, selected companion reads, actual private passage reads, and primary literature accesses. Public byte comparisons authenticate the assigned source against the stated commit. The independent source reviewer did not certify the source's companion RG argument, preliminary strict mixing gain, or continuum reconstruction. Those limitations remain material. The source has not been declared either globally proved or refuted by this analysis. Successful LaTeX compilation is solely a document check.

Private passage locators, relative to the user's Documents directory, are `RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex` (physical norm, projection and full-residual proofs), `RamCalBooks/volumes/volume21/chapters/ch02/finite.tex` (the finite positive-moment chapter), and `RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex` (lines 1–72, finite local theorem and remainder proof). The earlier author also read the Volume 21 spectral-limits chapter; the conclusions used here do not rely on an unquoted new theorem from it. The accompanying provenance file records exact local hashes and the narrower companion ranges. Inventories are discovery aids, not substitutes for reading the cited mathematical statements.

12 Future Research With Explicit Targets

A useful next computational project is a positive-window certificate that uses interval heat-trace or field-moment data, including zeroth moments. It should report the polynomial approximation error and coefficient-weighted data errors separately, and test positivity on the whole window rather than only at sampled points. A second project is a domain-correct transfer trial basis with a certified physical Gram lower bound and an explicit omitted-residual norm. These are finite, falsifiable goals; no result here promises a uniform continuum cost.

The analytic priorities are harder. One would need quantitative uniform moment-convergence rates to replace the order-of-limits existence argument by usable width/cutoff bounds. Control at the two-excitation threshold requires accounting for pairs of nonvacuum states rather than the vacuum-pair recursion. Scattering or asymptotic completeness would require additional multipoint and sector information absent from a two-point mass-support argument. Finally, a full independent validation of the continuum construction is a separate task. Improving a finite spectral certificate does not silently complete it.

13 Conclusion and Contribution Assessment

Ram Calculus is relevant here as a disciplined moment and physical operator representation, with remainders, positivity and transported metrics retained. The source's core infinite-volume difficulty is not removed by that representation. The proved additions are the fixed-window subthreshold count and mass-support extensions, plus explicit noise, residual-visibility and conditional residue bounds. They make several steps more controllable and adaptable without claiming a new particle construction. Accordingly, L4/5 applicability coexists with C2/5 realized contribution. No numerical mass, residue, historical novelty, global proof verification, or speedup is claimed.

References

[1] OpenAI. *An Isolated Particle Pole for the Two-Dimensional $O(3)$ Spin Field*. October 4, 2026. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/An-Isolated-Particle-Pole-for-the-Two-Dimensional-O3-Spin-Field-October-4-2026. Actual mathematical source, Sections 1–7.

[2] OpenAI. *The canonical massive continuum limit of the two-dimensional $O(3)$ model*. October 4, 2026. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-canonical-massive-continuum-limit-of-the-two-dimensional-O3-model-October-4-2026. Selected trace, moment, and reconstruction passages inspected; not an independent verification of the full construction.

[3] M. Subbarao. *Ram Calculus*, Volume 21, Chapter 1, physical-metrics fragment. Private local corpus; full residual and nonorthogonal physical norm statements actually read.

[4] M. Subbarao. *Ram Calculus*, Volume 21, Chapter 2, finite positive moments. Private local corpus; finite atomic certificates and nonidentifiability boundaries actually read.

[5] M. Subbarao. *Ram Calculus*, Volume 1, revised Chapter 3. Private local corpus; finite local moment identity and integral remainder theorem, not an assumed infinite expansion.

[6] L. Köhler-Schindler and V. Tassion. Crossing probabilities for planar percolation. *Duke Mathematical Journal* 172 (2023), 809–838. https://arxiv.org/abs/2011.04618. The source uses the preprint theorem; its contract was checked, not the full independent RSW proof.

[7] Y. Tokushige. Graphical proof of Ginibre's inequality. *Journal of Statistical Physics* 192 (2025), 25. https://doi.org/10.1007/s10955-024-03378-6. Primary theorem and nearby definitions checked.

[8] J. Balog and M. Niedermaier. Off-shell dynamics of the $O(3)$ nonlinear sigma-model beyond Monte Carlo and perturbation theory. *Nuclear Physics B* 500 (1997), 421–461. https://arxiv.org/abs/hep-th/9612039. Selected introductory and Section 2.1 premises checked by the independent reviewer, not a full form-factor proof audit.

[9] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association* 58 (1963), 13–30. https://doi.org/10.1080/01621459.1963.10500830. Classical concentration input used by the source.

[10] OpenAI. Public mathematics release revision history, October 7 entry, checked October 8, 2026. https://github.com/openai/math/blob/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/history.md.